Functions, Derivatives and Integrals

Mathematical Foundation

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Agenda

1. Introduction to basic terminologies
2. Functions
3. Calculus
4. Limits
5. Derivatives
   a. Rules in derivatives
   b. Partial differentiation
   c. Application of derivatives
6. Jacobian and Hessian matrix

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Introduction to basic terminologies

Interval

Given two real numbers a and b with $a < b$,
Closed Interval: The closed interval [a, b] is defined as the set of all real numbers x such that $a \leq x$ and $x \leq b$, or more concisely, $a \leq x \leq b$.

Open Interval: The open interval (a, b) is defined as the set of all real numbers x such that $a < x < b$.

a is called the lower endpoint, and b the upper endpoint.

The endpoints a and b are elements of [a, b]

but are not elements of (a, b).

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Examples of Intervals

Type	Symbol	Defining Formula
Closed	$[a, b]$	$a \le x \le b$
Open	(a, b)	$a < x < b$
Open	(a, ∞)	$a < x$
Open	$(-\infty, b)$	$x < b$
Open	$(-\infty, \infty)$	
Half-open	$[a, b)$	$a \le x < b$
Half-open	$[a, \infty)$	$a \le x$
Half-open	$(a, b]$	$a < x \le b$
Half-open	$(-\infty,b$	$x \le b$

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Boundary and Interior point

The set of interior points of the interval $[a, b]$ is (a, b), and the boundary points are a and b.

The set of interior points of the interval (a, b) is (a, b), and the boundary points are a and b.

Note: the boundary points of (a, b) are *not* members of the set (a,b).

- The <u>interval</u> I is **open** if every point in I is an <u>interior point</u> of I.
- The <u>interval</u> I is **closed** if every <u>boundary point</u> of I is a member of I.

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n-vector

Sets of *n*-tuples of numbers or an ordered list of *n* numbers is called an ***n*-vector**.

If n is greater than or equal to two, vectors are sometimes considered synonymous with points and so *n*-tuples (x_1, x_2,...,x_n) are sometimes called points in *n*-space.

Example:

An example of such a set for $n = 2$ is the set of pairs ("2-tuples") ($x1$, $x2$) such that $0 \leq x1 \leq 1$ and $0 \leq x2 \leq 1$, which is the set of points in a square with side of length 1 and bottom left corner at the origin.

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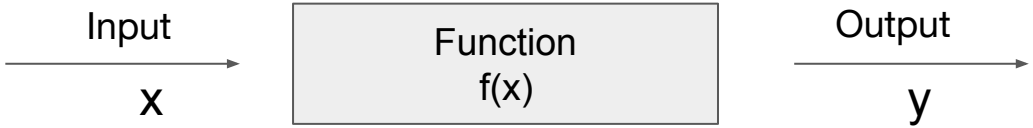
Function

A function is a relationship between input and output.



A **function** is a rule that associates with every member of some set, a single member of another set. The first set is called the **domain** of the function. A function with domain A is said to be **defined on** A.

Examples:

- Domain: $[-2, 1]$. Rule: $f(x) = 2x$.
- Domain: $(-1, 1)$. Rule: $f(x) = x$ if $x \geq 0$; $f(x) = 1/x$ if $x < 0$.

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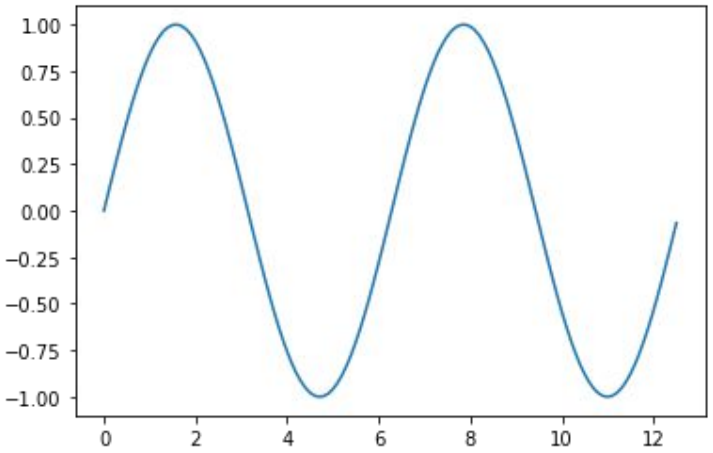
Function Notation

$f : R^D \rightarrow R$ specifies that f is a mapping from R^D to R

$x \rightarrow f(x)$ to specify a function, specifies the explicit assignment of an input x to a function value f(x).

A function f assigns every input x exactly one function value f(x)



Plot for sin function

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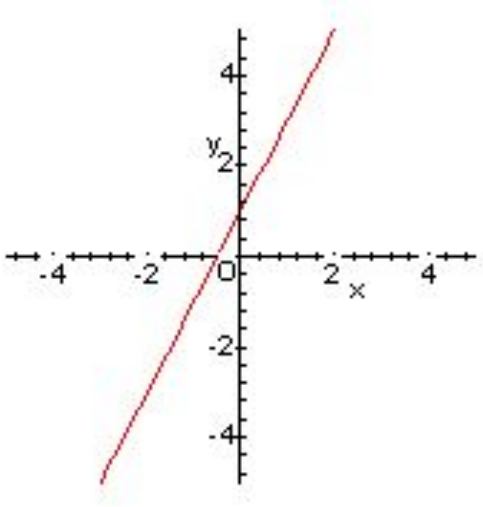
Graph of a function

Consider the function f(x) = 2 x + 1. We recognize the equation y = 2 x + 1 as the Slope-Intercept form of the equation of a line with slope 2 and y-intercept (0,1).



Think of a point moving on the graph of f. As the point moves toward the right it rises. This is what it means for a function to be *increasing*

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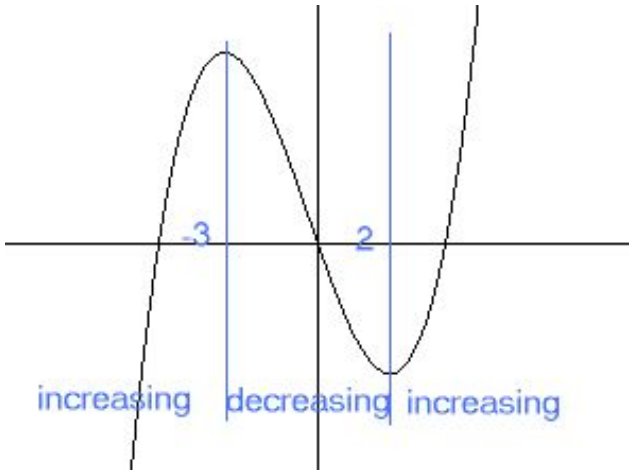
Increasing and Decreasing Functions

The function graphed above is decreasing for x between -3 and 2. It is increasing for x less than -3 and for x greater than 2.

Using interval notation, we say that the function is

decreasing on the interval (-3, 2)

increasing on (-infinity, -3) and (2, infinity)



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Calculus

- Gottfried Leibniz, developed the fundamentals of calculus independently
- Calculus is the study of how the functions change with respect to their input variables over time and it allows you to investigate and manipulate them.

Comprises of

- Differential Calculus and
- Integral Calculus

Applications: find velocity, area under the curve, optimisation, machine learning and deep learning

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Limit theorem

We introduce the idea of limit by considering the below problem

Suppose an object moves along the x-axis;

How to “find” the velocity of an object at a particular instant ?

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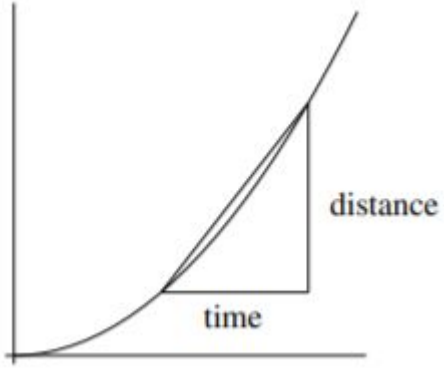
Find the velocity

Suppose an object moves along the x-axis and its displacement (in meters) s at time t (in seconds) is given by

$$s(t) = t^2, \qquad t \geq 0.$$

We want to consider its velocity at a certain time instant, say at t = 2. Idea
Velocity (or speed) is defined by

$$\text{velocity} = \frac{\text{distance traveled}}{\text{time elapsed}}$$



But, this can only be applied to find average velocities over time intervals. We (still) don't have a definition for velocity at t = 2.

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n	Time interval	Velocity
1	[2,2.5]	4.5 m/s
2	[2,2.25]	4.25 m/s
3	[2,2.125]	4.125 m/s
4	[2,2.0625]	4.0625 m/s
⋮		

The velocity v_n over the time interval $[2, 2+ 1/2^n]$ is

$$v_n = ((2 + 1/2^n)^2 - 2^2) / (1/2^n)$$

$$= ((4 + 2 * 2* (1/2^n) + (1/2^n)^2 - 4) / (1/2^n)$$

$$= 4 + (1/2^n)$$

From the above, it can be seen that if n is very large (ie., if the time interval is very short), v_n is very close to 4.

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Derivatives

In the example discussed, velocity is a function of time.

$$f(t) = v_n \text{ at time } t$$

If we have to determine the rate of change of the position of object at time t, we can do so, by taking the derivative of the function f(t) w.r.t t.

- Derivative refers to a rate of change of a function
- It is denoted as f'(t) or df/dt
- It is a signed quantity, with the sign indicating the direction of motion -- toward or away from the chosen origin.
- The absolute value of the velocity, |f'(t)|, is the speed of the object, which reflects how quickly it is moving regardless of direction.

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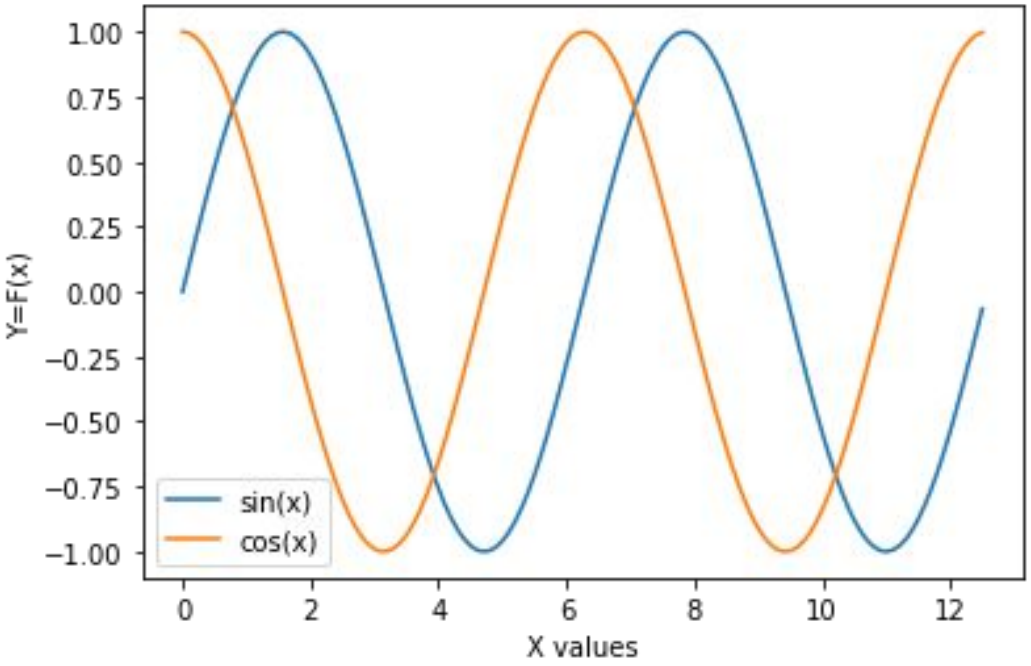
Sine function and Derivative



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Slope

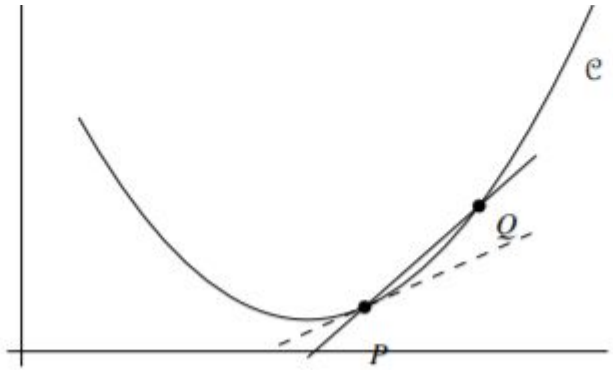
Consider the curve shown. The slope changes as we move along the curve.

At Q, the slope is very steep, whereas at P, the slope is gentle.

If the coordinates of P and Q are (x_1,y_1) and (x_2,y_2), then

Slope of the line joining the points P and Q is given by

$$\frac{y_2 - y_1}{x_2 - x_1}$$



The line PQ is called a secant line at P.
The slope is denoted as m_{PQ} and it changes as the point Q changes on the curve.

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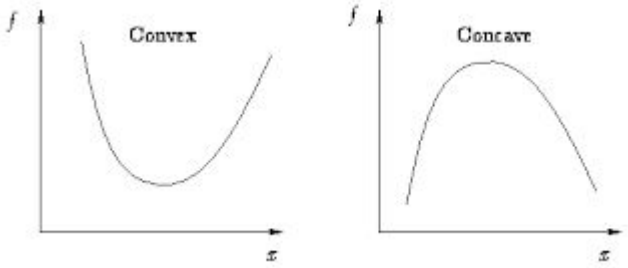
Concave and Convex Functions

Convex

A **function** that is **convex** is often synonymously called **concave** upwards, and a **function** that is **concave** is often synonymously called **concave** downward.



Concave

A differentiable **function** f is **concave** on an interval if its derivative **function** f′ is decreasing on that interval: a **concave function** has a decreasing slope.

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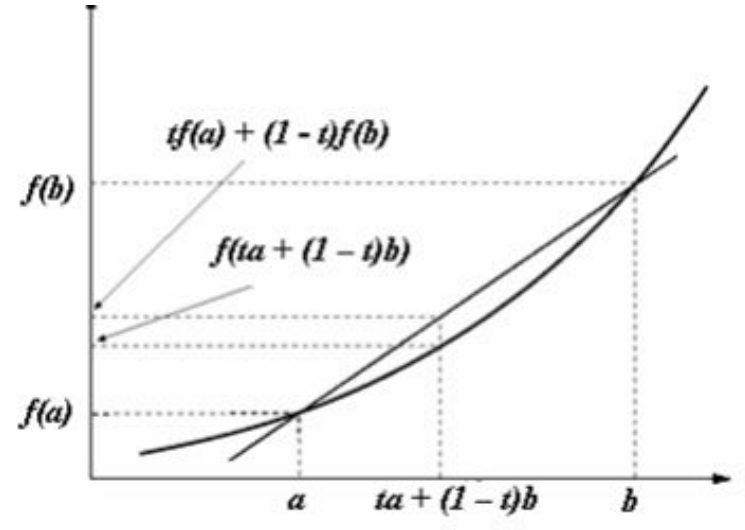
Convex Function

A real-valued function f defined on an interval (or on any convex subset C of some vector space) is called convex, if for any two points a and b in its domain C and any t in [0,1], we have

$$f(ta + (1-t)b) \leq t\,f(a) + (1-t)\,f(b)$$



A continuous function on an interval C is convex if and only if

$$f\left(\frac{a+b}{2}\right) \leq \frac{f(a)+f(b)}{2}$$

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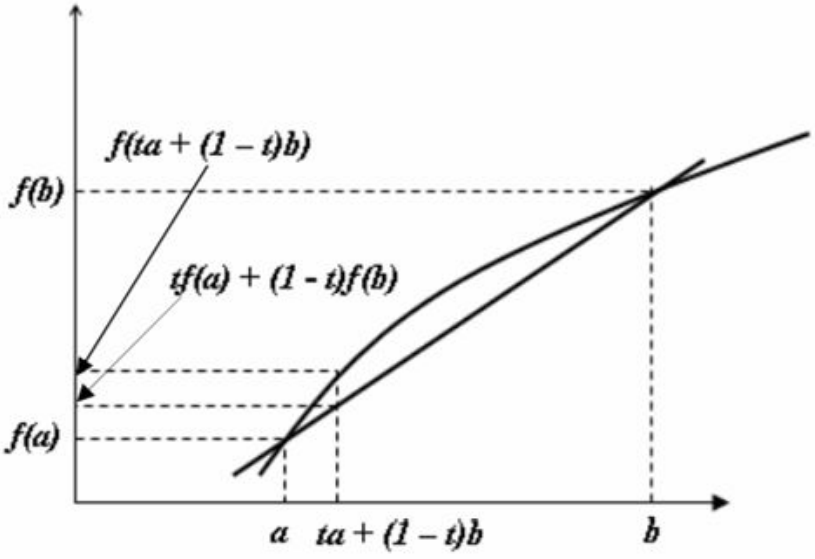
Concave Function

A function f(x) is said to be concave on an interval if, for all a and b in that interval,

$$\forall t \in [0,1], f(ta+(1-t)b) \geq tf(a)+(1-t)f(b)$$

A continuous function on C is concave if and only if

$$f\left(\frac{a+b}{2}\right) \geq \frac{f(a)+f(b)}{2}$$



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Slope and Derivative as Limit

The derivative of function f at x=c is the limit of the slope (m) of the secant line from x=c to x=c+h as h approaches 0.

$$m = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

Therefore, the slope of the tangent is the limit of Δy/Δx as Δx approaches zero, or dy/dx. We call this limit the derivative

$$m = \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

Differentiation is the process of finding the derivative of a function.

The differential, dF, of any differentiable function F is an infinitely small increment or change in the value of F. dF is measured in the same units as F itself

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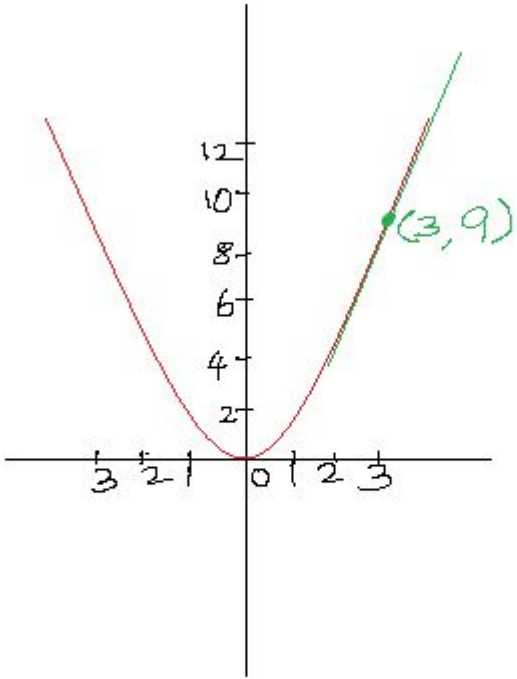
Example Find the slope of the curve given by $y = x^2$ at the point $P(3, 9)$.

Solution Put $f(x) = x^2$. By (4.1.1), the required slope (denoted by m_P) is

$$
\begin{aligned}
m_P &= \lim_{h \to 0} \frac{f(3+h) - f(3)}{h} \\
&= \lim_{h \to 0} \frac{(3+h)^2 - 3^2}{h} \\
&= \lim_{h \to 0} \frac{(9 + 6h + h^2) - 9}{h} \\
&= \lim_{h \to 0} \frac{6h + h^2}{h} \\
&= \lim_{h \to 0} (6 + h) \\
&= 6.
\end{aligned}
$$



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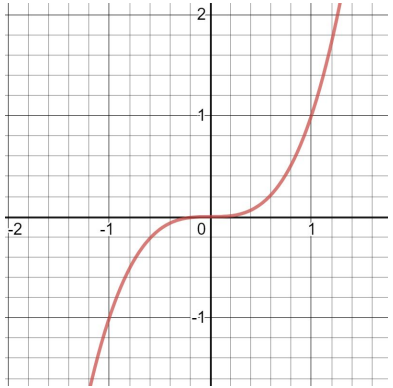
Example Let $f(x) = x^3$. Find $f'(x)$.

Solution By definition, we have

$$
\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{(x+h)^3 - x^3}{h} \\
&= \lim_{h\to 0} \frac{(x^3 + 3x^2h + 3xh^2 + h^3) - x^3}{h} \\
&= \lim_{h\to 0} \frac{3x^2h + 3xh^2 + h^3}{h} \\
&= \lim_{h\to 0} (3x^2 + 3xh + h^2) \\
&= 3x^2.
\end{aligned}
$$



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Necessary conditions for Differentiation

Differentiability

To be differentiable at a given point:

- The function must be *continuous* at that point.
- The tangent line at that point cannot be vertical
- The line must be *smooth* at that point

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Derivatives of constant and identity functions

The derivation of a constant function is 0 (the zero function), that is

$$\frac{d}{dx}c = 0,$$

The derivation of the identity function is the constant function, that is

$$\frac{d}{dx}x = 1.$$

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Rules for Differentiation

Power Rule (positive integer version) : Let n be a positive integer. Then the power function x^n is differentiable on R and we have

$$\boxed{\frac{d}{dx}x^n = nx^{n-1}}$$

Constant Multiple Rule: Let f be a function and let k be a constant. Suppose that f is differentiable. Then the function kf is also differentiable. Thus we have,

$$\boxed{\frac{d}{dx}(kf)(x) = k \cdot \frac{d}{dx}f(x)}$$

Sum Rule: Let f and g be functions with the same domain. Suppose that f and g are differentiable. Then the function f + g is also differentiable. Thus we have,

$$\boxed{\frac{d}{dx}(f+g)(x) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x).}$$

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Chain rule

Chain Rule deals with nested functions.

The derivative is calculated as the derivative of an outer function multiplied with inner function, then all multiplied by the derivative of the inner function.

$$\boldsymbol{F(x) = f\big(g(x)\big) \quad \rightarrow \quad F'(x) = f'\big(g(x)\big) \times g'(x)}$$

And here's a simple example calculated by hand:

$$\boldsymbol{F(x) = \left(x^2 - 3x + 5\right)^3}$$

$$F'(x) = 3 \times (x^2 - 3x + 5)^2 \times (2x - 3)$$

$$F'(x) = (6x - 9)(x^2 - 3x + 5)^2$$

Python Code

```
x = sym.Symbol('x')
sym.diff(((x**2)-3*x+5)**3)
(6*x - 9)*(x**2 - 3*x + 5)**2
```

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Product Rule

Product Rule *Let f and g be functions with the same domain. Suppose that f and g are differentiable. Then the function fg is also differentiable. Moreover, we have*

$$\frac{\mathrm{d}}{\mathrm{d}x}(fg)(x) = g(x) \cdot \frac{\mathrm{d}}{\mathrm{d}x}f(x) + f(x) \cdot \frac{\mathrm{d}}{\mathrm{d}x}g(x).$$

$$\boldsymbol{F(x) = f(x) \times g(x) \quad \rightarrow \quad F'(x) = f(x) \times g'(x) + g(x) \times f'(x)}$$

Let's calculate one example by hand. We have the following:

$$\boldsymbol{F(x) = \left(x^2 + 1\right) \times \cos(x)}$$

$$F'(x) = (x^2 + 1) \times (-\sin(x)) + \cos(x) \times (2x)$$

$$F'(x) = 2x\cos(x) - (x^2 + 1)\sin(x)$$

Python Code

```
x = sym.Symbol('x')
sym.diff((x**2+1) * sym.cos(x))
2*X*cos(X) - (X**2 + 1)*sin(X)
```

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Properties of Differentials

For any differentiable function F(x),

1. $dF = F'(x)dx$ (Recall that $F'(x) = dF/dx$!).
2. For any constant C, $dC = 0$.
3. For any constant C and differentiable function F(x),

   $d(CF) = CdF = CF'(x)dx$.

4. For any differentiable functions u and v, $d(u \pm v) = du \pm dv$.

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Univariable and Multivariable calculus

- Univariate calculus deals with single variables in a function

  Example: The function $f(x) = x^2 + 2x - 1$ is a univariate function

- Multivariate calculus deals with two or more variables in a function

  Example: The function $f(x,y) = 2x + y - 1$ is a multivariate function

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Higher-Order Derivatives

First order derivative:

Let f be a function that is differentiable at some points belonging to dom(f). Then f' is a function which is the first order derivative.

Second order derivative:

If, in addition, f' is differentiable at some points belonging to dom(f'), then the derivative of f' exists and is denoted by f''.

It is the function given by $$f''(x) = \lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h}$$

Third derivative:

If f" is differentiable at some points belonging to dom(f"), then the derivative of f" is denoted by f'''' called the third derivative of f.

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Partial Derivatives

A partial derivative of a function of several variables is its **derivative with respect to one of those variables**, with the others held constant.

Consider $f(x, y) = x^3 + 2x^2 y + y^2 + 2x + 1$

Suppose we keep y constant and vary x; then what is the rate of change of the function f?

Differentiating w.r.t. x we get: $3x^2 + 12x + 2$.

We say that f has been partially differentiated with respect to x.

We denote the partial derivative of f with respect to x by $\partial f / \partial x$ (to be read as 'partial dee f by dee x'). In this example, when y = 3:

$$\partial f / \partial x = 3x^2 + 12x + 2$$

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Partial differential equation

A partial differential equation (PDE) is a differential equation that contains unknown multivariable functions and their partial derivatives. PDEs are used to formulate problems involving functions of several variables

A first order PDE in two independent variables x, y and the dependent variable z can be written as

$$f(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}) = 0. \tag{1}$$

For convenience, we set

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}.$$

Equation (1) then takes the form

$$f(x, y, z, p, q) = 0. \tag{2}$$

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Jacobian Matrix

The matrix containing all of the partial derivatives of a function (f) whose input and output are both vectors is the Jacobian matrix.

$$\mathbf{f}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} f_1(x,y) \\ f_2(x,y) \end{bmatrix} = \begin{bmatrix} x^2 y \\ 5x + \sin y \end{bmatrix}.$$

$$\mathbf{J_f}(x,y) = \begin{bmatrix} \dfrac{\partial f_1}{\partial x} & \dfrac{\partial f_1}{\partial y} \\ \\ \dfrac{\partial f_2}{\partial x} & \dfrac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} 2xy & x^2 \\ 5 & \cos y \end{bmatrix}$$

Python code

```
from sympy import sin, cos, Matrix


x,y = sym.symbols('x,y')
Fxy = sym.Matrix([(x**2)*y,  5*x + sym.sin(y)])
Fxy.jacobian([x,y])
```

Jacobian Matrix

```
Matrix([
[2*x*y,   x**2],
[    5, cos(y)]])
```

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Hessian matrix

1. Hessian is a square matrix of second order partial derivatives of a scalar-valued function or scalar field.
2. It describes the local curvature of a function of many variables.
3. Hessian matrix here is always symmetric

The Hessian matrix of n variables is given by

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1\,\partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1\,\partial x_n} \\ \frac{\partial^2 f}{\partial x_2\,\partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2\,\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n\,\partial x_1} & \frac{\partial^2 f}{\partial x_n\,\partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

$f(x,y) = -7x^2 - 5xy - 4y^2$

Python Code

```
from sympy.abc import x, y
from sympy import ordered, Matrix, hessian
x,y = sym.symbols('x,y')
eq = -7*(x**2) - 5*x*y - 4*(y**2)
v = list(ordered(eq.free_symbols))
hessian(eq,v)
```

```
H =   Matrix([
              [-14, -5],
              [ -5, -8]])
```

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Applications of Derivatives

- Rate of Change of Quantities
- Approximations
- Increasing and Decreasing Functions
- Maxima and Minima
- Tangents and Normals

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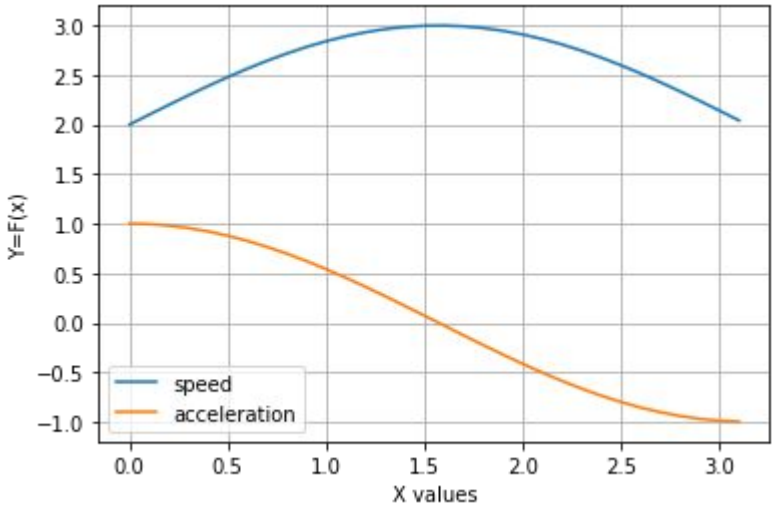
Finding acceleration from speed

A vehicle has initial speed of 2 km/hr and the reaches maximum speed of 3 km/hr. After the maximum speed, the vehicle slows down decelerate back to the initial speed. The blue line represents the vehicle trajectory

Then the acceleration can be represented by the redline as the derivative of the speed



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Derivatives for images

$$f'(x) \approx \frac{f(x+0.5h) - f(x-0.5h)}{h}$$

At each image point, the gradient vector points in the direction of largest possible intensity increase, and the magnitude corresponds to the rate of change in that direction. Thus for an image f(x,y), the gradient direction and magnitude is given by

Magnitude : $$||\nabla f|| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

Direction : $$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$

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Gradients and image processing

The gradient of an N-variable function at each point is an N-D vector with the components given by the derivatives in the N-directions. e.g. for a 3-variable function (f(x,y,z)), the gradient, if it exists, is given by

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$$

Thus, the gradient provides two pieces of information – magnitude and direction. The direction of the gradient tells us the direction of greatest increase while the magnitude represents the rate of increase in that direction.

Because gradients are defined only for continuous functions and Image is a 2-d **discrete** function (F(x,y)). Thus we need to approximate the gradients and we do this using Finite differences.

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Appendix

```
x = sym.Symbol('x')
y = sym.Symbol('y')
```

Expand a function

```
sym.expand((x + y) ** 3)                    # x**3 + 3*x**2*y + 3*x*y**2 + y**3
```

Simplify a function

```
sym.simplify((x + x * y) / x)               # y + 1
```

Limit

```
sym.limit(sym.sin(x) / x, x, 0)
```

Derivative

```
sym.diff(sym.sin(x), x)
```

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Higher derivatives can be calculated using the diff(func, var, n) method

```
sym.diff(sym.sin(2 * x), x, 1)          # 2*cos(2*x)
```

Chain Rule:

```
x = sym.Symbol('x')
sym.diff(((x**2)-3*x+5)**3)             # (6*x - 9)*(x**2 - 3*x + 5)**2
```

Product Rule

```
x = sym.Symbol('X')
sym.diff((x**2+1) * sym.cos(x))         # 2*X*cos(X) - (X**2 + 1)*sin(X)
```

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Thank you!

Happy Learning :)

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